

1. Which of the following costs would you consider when making a capital budgeting decision?

- A) Sunk cost
- B) Opportunity cost**
- C) Interest expense
- D) Fixed overhead cost

2. Temporary Housing Services Incorporated (THSI) is considering a project that involves setting up a temporary housing facility in an area recently damaged by a hurricane. THSI will lease space in this facility to various agencies and groups providing relief services to the area. THSI estimates that this project will initially cost \$5 million to set up and will generate \$20 million in revenues during its first and only year in operation (paid in one year). Operating expenses are expected to total \$12 million during this year and depreciation expense will be another \$3 million. THSI will require no working capital for this investment. THSI's marginal tax rate is 35%. Assume that THSI's cost of capital for this project is 15%. The NPV of this temporary housing project is closest to:

- A) \$435,000**
- B) -\$650,000
- C) \$1,960,000
- D) -\$435,000

ANSWER:

$$NI = (20 - 12 - 3) * (1 - 0.35) = 3.25$$

$$FCF = 3.25 + 3 = 6.25$$

$$NPV = -5 + 6.25 / 1.15 = 0.43478$$

3. Which of the following statements is FALSE?

- A) Fluctuations of a stock's returns that are due to firm-specific news are common risks.**
- B) The volatility in a large portfolio will decline until only the systematic risk remains.
- C) When we combine many stocks in a large portfolio, the firm-specific risks for each stock will average out and be diversified.
- D) The risk premium of a security is determined by its systematic risk and does not depend on its diversifiable risk.

4. Consider a portfolio that consists of an equal investment in 20 firms. For each of these firms, there is a 70% probability that the firms will have a 16% return and a 30% that they will have a -8% return. Each of these firms' returns is independent of all others. The standard deviation of this portfolio is closest to:

- A) 2.5%**
- B) 4.2%
- C) 8.8%
- D) 11.0%

ANSWER:

$$E(R) = (30\% * (-8)) + (70\% * 16) = 8.8$$

$$\text{For 1 firm : } Sd = ((-8 - 8.8)^2 * 30\% + ((16 - 8.8)^2 * 70\%)^{0.5} = 10.998$$

$$\text{For 10 firms: } 10.998 / (20^{0.5}) = 2.459$$

5. Which of the following statements is TRUE?

- A) The risk premium of a zero-beta stock should be higher than the market risk premium
- B) The risk premium of a zero-beta stock is equal to the market risk premium
- C) The risk premium of a zero-beta stock is negative
- D) The risk premium of a zero-beta stock is zero**

6. Suppose Autodesk stock has a beta of 2.16, whereas Costco stock has a beta of 0.69. If the riskfree interest rate is 4% and the expected return of the market portfolio is 10%, what is the expected return of a portfolio that consists of 60% Autodesk stock and 40% Costco stock, according to the CAPM?

- A) 1.57
- B) 12.55
- C) 13.43**
- D) 19.72

ANSWER:

$$E(R)_{\text{Autodesk}} = 4 + 2.16 \cdot (10 - 4) = 16.96$$

$$E(R)_{\text{Costco}} = 4 + 0.69 \cdot (10 - 4) = 8.14$$

$$E(R)_{\text{portfolio}} = 16.96 \cdot 60\% + 8.14 \cdot 40\% = 13.43$$

7. Why the debt cost of capital is less than its yield to maturity?

- A) Because of the systematic risk
- B) Because default risk can be diversified
- C) Because yield to maturity is the debt's expected return
- D) Because yield to maturity includes the default risk premium**

8. Suppose XYZ stock has a beta of 0.6. The risk-free rate is 5% and the expected return of the market portfolio is 8%. XYZ is 25% debt financed, and you anticipate its debt cost of capital will be 6%. If its corporate tax rate is 0%, what is your estimate of its WACC?

- A) 6%
- B) 6.6%**
- C) 6.8%
- D) 7%

ANSWER:

$$E(R) = 5 + 0.6 \cdot (8 - 5) = 6.8$$

$$WACC = 6.8 \cdot 75\% + 6 \cdot 25\% = 6.60$$

9. Which of the following statements is FALSE?

- A) Leverage decreases the risk of the equity of a firm.**
- B) Because the cash flows of the debt and equity sum to the cash flows of the project, by the Law of One Price the combined values of debt and equity must be equal to the cash flows of the project.
- C) Franco Modigliani and Merton Miller argued that with perfect capital markets, the total value of a firm should not depend on its capital structure.
- D) It is inappropriate to discount the cash flows of levered equity at the same discount rate that we use for unlevered equity.

10. Consider two firms, With and Without, that have identical assets that generate identical cash flows. Without is an all-equity firm, with 1 million shares outstanding that trade for a

price of \$24 per share. With has 2 million shares outstanding and \$12 million dollars in debt at an interest rate of 5%. According to MM Proposition 1, the stock price for With is closest to:

- A) \$8.00
- B) \$24.00
- C) \$6.00**
- D) \$12.00

ANSWER:

*Market value of equity of WITHOUT=Market value of firm=1*24=24*

Market value of equity of WITH=24-12=12

Price=12/2=6

11. Which of the following statements is FALSE?

- A) In general, the gain to investors from the tax deductibility of interest payments is referred to as the interest tax shield.
- B) The interest tax shield is the additional amount that a firm would have paid in taxes if it did not have leverage.
- C) Because Corporations pay taxes on their profits after interest payments are deducted, interest expenses reduce the amount of corporate tax firms must pay.
- D) As Modigliani and Miller made clear in their original work, capital structure matters in perfect capital markets. Thus, if capital structure does not matter, then it must stem from a market imperfection.**

12. KD Industries has 30 million shares outstanding with a market price of \$20 per share and no debt. KD has had consistently stable earnings, and pays a 35% tax rate. Management plans to borrow \$200 million on a permanent basis through a leveraged recapitalization in which they would use the borrowed funds to repurchase outstanding shares. After the recapitalization, the value of KD's levered equity is closest to:

- ~~A) \$900 million~~
 - B) \$780 million**
 - ~~C) \$980 million~~
 - ~~D) \$700 million~~
- A) \$670 million
 - B) \$400 million
 - C) \$330 million
 - D) \$470 million**

ANSWER:

*MV of eq.=30*20+200*0.35-200=470*

13. Which of the following statements is FALSE?

- A) The tradeoff theory weighs the costs of debt that result from shielding cash flows from taxes against the benefits from the effects of financial distress associated with leverage.**
- B) Leverage has costs as well as benefits.
- C) According to the tradeoff theory, the total value of a levered firm equals the value of the firm without leverage plus the present value of the tax savings from debt, less the present value of financial distress costs.

- D) Firms have an incentive to increase leverage to exploit the tax benefits of debt. But with too much debt, they are more likely to risk default and incur financial distress costs.

14. Luther Industries has no debt and expects to generate free cash flows of \$48 million each year. Its cost of capital 16%. Luther believes that if it permanently increases its level of debt to \$100 million, the risk of financial distress may cause it to lose some customers and receive less favorable terms from its suppliers. As a result, Luther's expected free cash flows with debt will be only \$44 million per year. Suppose Luther's tax rate is 40%. The value of Luther with leverage is closest to:

- A) **\$315 million**
- B) \$340 million
- C) \$205 million
- D) \$300 million

ANSWER:

$$V = 44 / 0.16 + 100 * 0.40 = 315$$

15. Which of the following statements is FALSE?

- A) **Unlike with capital structure, taxes are not an important market imperfection that influence a firm's decision to pay dividends or repurchase shares.**
- B) If dividends are taxed at a higher rate than capital gains, which has been true until the most recent change to the tax code, shareholders will prefer share repurchases to dividends.
- C) Shareholders typically must pay taxes on the dividends they receive. They must also pay capital gains taxes when they sell their shares.
- D) Because long-term investors can defer the capital gains tax until they sell, there is still a tax advantage for share repurchases over dividends.

16. The JRN Corporation will pay a constant dividend of \$3 per share, per year, in perpetuity. Assume that all investors pay a 20% tax on dividends and that there is no capital gains tax. The cost of capital for investing in JRN stock is 12%. The price of a share of JRN's stock is closest to:

- A) **\$20.00**
- B) \$24.00
- C) \$25.00
- D) \$18.00

ANSWER:

$$P = 3 * (1 - 0.2) / 0.12 = 20$$

17. Assume that France, one of the European countries that uses the euro as its currency, would prefer that its currency depreciate against the dollar. Can it apply central bank intervention to achieve this objective?

- A) Yes
- B) **No**
- C) Maybe
- D) There is not any central bank that can influence the FX rate of own currency.

18. To increase the value of its home currency, a central bank could attempt to increase interest rates, thereby attracting a foreign demand for the home currency to buy high-yield securities.

How can this intervention be called?

- A) **Indirect intervention**
- B) Direct intervention
- C) Sterilized intervention
- D) It does not belong to the intervention.

19. Assume the following information:

	<u>Beal Bank</u>	<u>Yardley Bank</u>
Bid price of New Zealand dollar	\$.401	\$.398
Ask price of New Zealand dollar	\$.404	\$.400

Given this information, is locational arbitrage possible? If so, explain the steps involved in locational arbitrage, and compute the profit from this arbitrage if you had \$1,000,000 to use. How much is the profit?

- A) **\$2,500**
- B) \$-14,500
- C) \$15,075
- D) \$10,000

ANSWER: Yes! One could purchase New Zealand dollars at Yardley Bank for \$.40 and sell them to Beal Bank for \$.401. With \$1 million available, 2.5 million New Zealand dollars could be purchased at Yardley Bank. These New Zealand dollars could then be sold to Beal Bank for \$1,002,500, thereby generating a profit of \$2,500.

20. Purchasing power parity (PPP) theory attempts to quantify this inflation – exchange rate relationship. There is one of the following statements that is not correct:

- A) The absolute form of PPP is an extension of the law of one price.
- B) **The absolute form of PPP suggests that the prices of the same products in different countries should be different when measured in a common currency.**
- C) The relative form of PPP accounts for market imperfections like transportation costs, tariffs, and quotas. It states that the differences should be constant.
- D) The government agencies use it to compare the output of countries that use different exchange rates.

21. Please select the correct answer. The international Fisher effect (IFE) theory suggests that

- A) currencies with higher interest rates will appreciate because higher nominal rates is attractive for the investor

- B) investors hoping to capitalize on a higher foreign interest rate should earn a higher return on the investment but lose all the gain in the depreciation of the currency.**
- C) currencies with higher interest rates will appreciate because higher nominal rates reflect higher expected inflation.
- D) currencies with higher interest rates will depreciate because the real interest rate directly influence the export and the import.

22. Forecasting the Future Spot Rate Based on IFE. Assume that the spot exchange rate of the Singapore dollar is \$.70.

The one-year interest rate is 11 percent in the United States and 7 percent in Singapore. What will the spot rate be in one year according to the IFE?

Answer:

- A) \$ 0.7262**
- B) \$ 2.07
- C) \$ 0.68
- D) \$ 2.12

Answer: $ef = (1+ih)/(1+if) - 1 = (1+0,11)/(1+0,07) - 1 = 0,0374$

Spot rate : $S*(1+ef) = \$.70 \times (1 + .0374) = \$.7262$.

23. Segmented capital markets means firms may face differential access to markets, because

- A) In some countries, especially in the developed world, investors or companies do not have equal access to financial securities.
- B) if there is no any kind of asymmetry with respect to information about them.**
- C) **In the developing world, all companies do not have equal access to financial securities.**
- D) Segmented capital markets do not include capital controls and foreign exchange controls that create barriers to international capital flows, and thus segment national markets

24. Given the following information: The risk-free interest rate 4% in the domestic country. The cost of debt is 5.2%. The company's long term capital structure is 60% equity and 40% debt.

The expected rate of return of the domestic market is 9% and the international market is 8%.

The company systematic risk indicator (beta) is 1,2 in the domestic market and 0,9 in the international market.

What is the more favorable equity funding for this company based on this information.

- A) the domestic market, because the kwacc 8 %
- B) the international market, because the kwacc 6,64 %**

- C) the domestic market, because the kwacc 8,8% and the global is 11 %
- D) there is no different

25. Which of the following statistical indicator cannot have a value of -1?

- A) beta
- B) correlation
- C) mean
- D) **standard deviation**